Shailendra Dabral

MS-R Student in Astronomy, Astrophysics & Space Engineering
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Research Interests: Generative AI • Scientific Machine learning • Remote sensing

Education

Indian Institute of Technology (IIT) Indore — Indore, India
Master of Science - Research (MS-R), Astronomy & Space Engineering — *2024 - 2026*

- **Current Status:** Second-year student, Thesis submitted
- **Coursework:** General Relativity, Astrophysical Fluid Dynamics, Sensors and Detectors, Generative AI (CS Dept.), Algebraic Number Theory (Math Dept.).

Hemvati Nandan Bahuguna Garhwal University (Central University) — Uttarakhand, India
Master of Science in Physics — *2018 – 2020*

- **Coursework:** Advanced Quantum Mechanics, Electrodynamics & Astrophysics, Computational Physics, Particle Physics, Statistical Physics, Condensed Matter Physics.

Hemvati Nandan Bahuguna Garhwal University (Central University) — Uttarakhand, India
Bachelor of Science (Physics, Mathematics, Computer Science) — *2015 – 2018*

Publications & Presentations

- **Accepted:** "State Space Mamba-Based Change Detection Model," accepted for presentation at the *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*.
- **Published:** "GeoHam: Symplectic Hamiltonian Integration for Efficient Remote Sensing Change Detection," submitted to *IEEE Geoscience and Remote Sensing Letters (GRSL)* https://doi.org/10.1109/LGRS.2026.3728830.
- **Under Review:** "Amortized Chaotic System Identification and State Reconstruction via Physics-Informed Diffusion with Dormand–Prince Integration," evaluated on the Lorenz-63 system, submitted to *AIP Chaos*.
- Manuscript based on MS thesis detailing hybrid deep learning architectures for time-series change detection **Preprint:** DOI: 10.20944/preprints202605.1851.v1
- **Under Review:** "PIDM-CSS: A Conformal Symplectic Extension of Physics-Informed Diffusion for Multi-System Chaotic Reconstruction," extending the above across the Lorenz-63, Lorenz-96, Rössler, Rabinovich, and Hyper5D systems (course project under the Department of Computer Science), submitted to *Physica D: Nonlinear Phenomena*. **Preprint:** arXiv, DOI: 10.48550/arXiv.2605.26619
- **Workshops:** Actively participated in technical workshops including "Aditya L1", the "Square Kilometre Array (SKA)", and "Numerical Techniques for Atmospheric and Space Sciences".

Research Experience

Time Series Change Detection — IIT Indore
Research Assistant | Supervisor: Dr. Unmesh Khati (Remote Sensing) — *2024 – 2026*

- Developing a hybrid CNN and gradient-boosted (LightGBM) architecture for multi-temporal change detection that outperforms end-to-end baselines (U-Net, Siamese U-Net, ChangeFormer, STANet) on four-band and eight-band custom PlanetScope datasets with high class imbalance and limited data, while providing feature-importance-based interpretability that pure deep architectures lack.
- Conducting advanced time-series analysis on PlanetScope optical imagery to quantify spatio-temporal environmental changes.
- Developing novel deep learning architectures benchmarked against SOTA methods on public change-detection datasets (e.g., CLCD, OSCD, LEVIR-CD).

Generative AI in Physical Systems — IIT Indore
Interdisciplinary Project (Departments of Computer Science and Physics) — *2025 – 2026*

- Developing a physics-informed diffusion framework (PIDM-DP/PIDM-CSS) that combines diffusion models with Dormand–Prince and conformal symplectic integration to reconstruct trajectories of chaotic dynamical systems (Lorenz-63, Lorenz-96, Rössler, Rabinovich, Hyper5D).

Physics-Informed Neural Networks for Mars Atmospheric Data NASA Mars Mission
Current Research Work *2026 – Present*

- Developing physics-informed neural networks to reconstruct missing atmospheric data from sparse observations collected by NASA Mars missions.

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Technical Skills

Programming: Python, LaTeX, Julia.

Machine Learning: TensorFlow, PyTorch, Siamese Networks, U-Net Architectures, Vision Transformers, State Space Models (Mamba), Diffusion Models, Scientific Machine Learning, Generative AI.

Data Analysis: Statistical Analysis, Time Series Processing, Signal Processing.

Geospatial & Tools: GDAL, Rasterio, Google Earth Engine, QGIS

Operating Systems: Linux, Windows.

Professional Experience

Graduate Teaching Assistant IIT Indore
Department of Astronomy, Astrophysics & Space Engineering *2024 - 2026*

- **Current:** Served as T.A. for “Remote Sensing for Atmospheric Sciences” for Master’s students.
- **Previous:** Served as T.A. for “Electro-Magnetism” and “Quantum Mechanics” for first-year undergraduate students.

Awards

- Monthly research stipend, awarded after qualifying the highly competitive GATE examination.
- **Second Prize**, IEEE Hackathon Challenge.

Additional Skills & Information

- **Leadership:** Participated in multiple hackathons as Team Leader; served as Gym Coordinator for one year at IIT Indore.
- **Independent Initiatives (2020–2024):** Provided independent physics instruction; managed business operations for a café.
- **Clubs:** Core member of the Kalakriti (Drawing and Art) Club at IIT Indore.
- **Certifications:** Neural Networks and Deep Learning, Supervised Machine Learning, Improving Deep Neural Networks (Hyperparameter Tuning, Regularization, Optimization), Astrotech, Differential Equations for Engineers.
- **Languages:** English (Professional Working Proficiency), Hindi (Native).